

TSURUGA, Japan

WMO: 476310

Lat: 35.6533N

Lon: 136.0622E

Elev: 41

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 31.3	(c) 32.6	(d) 20.7	(e) 15.6	(f) 40.2	(g) 22.8	(h) 17.2	(i) 39.9	(j) 29.8	(k) 38.8	(l) 27.4	(m) 38.5	(n) 8.3	(o) 180	(p) 0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.0	(c) 92.8	(d) 77.2	(e) 90.9	(f) 77.1	(g) 88.8	(h) 76.7	(i) 79.4	(j) 88.2	(k) 78.3	(l) 87.1	(m) 77.5	(n) 86.0	(o) 12.0	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	140.4	84.5	75.9	135.7	83.8	75.0	131.3	83.1	43.0	88.3	41.9	87.3	41.1	86.2	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	22.2	19.7	27.9	96.3	2.0	1.4	26.5	97.3	25.4	98.2	24.3	99.0	22.9	100.0
			25.9	81.1	2.0	1.3	24.5	82.0	23.3	82.7	22.2	83.4	20.7	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	40.3	41.1	46.7	56.1	65.0	71.9	80.0	82.7	75.3	65.1	54.7	45.2
	DBStd	15.37	4.16	4.86	5.89	6.21	5.23	4.53	4.74	3.92	5.23	5.02	5.46	5.38
	HDD50	892	302	255	138	16	0	0	0	0	0	0	16	165
	HDD65	3331	766	670	567	275	66	2	0	1	60	311	613	
	CDD50	4706	1	5	36	198	464	658	930	1014	758	467	157	18
	CDD65	1668	0	0	0	6	65	210	465	549	309	62	2	0
	CDH74	14533	0	0	0	32	284	1044	4546	6206	2239	182	0	0
	CDH80	5053	0	0	0	3	36	204	1645	2562	592	11	0	0
Wind	WSAvg	9.2	10.1	10.4	10.3	9.8	9.4	8.2	8.1	8.3	8.1	8.3	9.1	9.8
Precipitation	PrecAvg	82.4	9.3	6.0	5.4	4.5	5.8	6.5	8.8	5.7	7.8	5.6	6.4	10.0
	PrecMax	109.0	17.0	9.0	8.9	7.4	18.0	11.9	17.9	15.3	16.5	11.3	12.8	21.5
	PrecMin	58.5	3.0	1.8	2.7	1.7	2.4	2.8	2.7	1.1	2.8	2.3	3.4	4.9
	PrecStd	10.8	3.5	1.7	1.7	1.7	3.5	2.3	3.9	3.2	3.8	1.9	1.9	3.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	60.1	68.3	78.4	83.9	88.1	94.8	95.7	91.5	81.6	71.6	62.8
		MCWB	48.7	52.8	54.6	59.6	65.9	73.1	77.2	77.1	75.7	70.3	60.7	55.5
	2%	DB	52.1	55.5	63.6	72.8	80.1	85.0	92.4	93.4	88.6	78.3	68.5	59.1
		MCWB	45.9	48.4	52.7	58.6	64.1	71.8	77.2	77.4	75.2	68.5	59.6	51.9
	5%	DB	49.7	52.4	59.9	69.6	77.2	82.2	90.3	91.7	85.9	75.8	66.1	56.6
		MCWB	43.7	46.1	51.1	58.1	63.1	70.7	77.1	77.5	74.4	66.3	58.1	50.1
	10%	DB	47.5	49.7	56.8	66.4	74.2	79.6	87.9	89.8	83.2	73.3	63.9	54.0
		MCWB	41.9	43.5	49.3	56.5	62.5	69.7	76.6	77.4	73.5	64.5	57.0	48.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	53.9	57.7	65.9	70.1	76.6	81.0	80.9	78.3	73.2	63.8	57.5
		MCDB	53.8	58.9	63.3	71.3	77.2	83.8	88.2	90.6	86.5	79.5	68.1	60.9
	2%	WB	46.8	49.7	55.0	61.8	67.3	74.7	79.2	79.7	76.9	70.2	61.7	53.3
		MCDB	50.6	54.1	61.3	68.9	74.9	81.4	87.6	89.1	84.8	76.3	66.2	57.6
	5%	WB	44.5	46.9	52.6	59.6	65.5	73.1	78.2	78.8	75.7	68.1	59.7	50.8
		MCDB	49.0	51.0	58.4	66.8	73.0	79.4	86.8	87.9	83.3	73.3	64.6	55.6
	10%	WB	42.5	44.3	50.1	57.6	64.1	71.5	77.3	78.0	74.6	66.1	57.8	48.5
		MCDB	46.7	49.1	56.1	64.1	71.8	77.5	85.8	86.8	81.8	71.8	63.2	53.4
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.7	13.2	14.2	13.8	11.3	11.0	12.0	11.9	12.9	12.3	10.5
		MCDBR	12.7	14.9	18.2	18.6	18.9	15.3	14.3	13.9	13.6	14.4	14.3	13.6
		MCWBR	8.8	10.6	11.7	9.5	8.6	6.9	5.4	5.3	5.8	7.1	8.7	9.1
	5% WB	MCDBR	11.4	12.7	14.5	14.3	14.1	11.5	12.7	12.7	11.9	11.1	11.4	12.0
		MCWBR	8.9	10.4	11.2	8.6	8.0	6.0	5.7	5.6	5.7	6.8	8.7	9.4
Clear-Sky Solar Irradiance	taub	0.437	0.469	0.534	0.560	0.564	0.587	0.581	0.552	0.498	0.461	0.438	0.434	
	taud	1.996	1.938	1.819	1.813	1.863	1.880	1.952	2.024	2.136	2.140	2.101	2.027	
	Ebn at Noon	228	237	233	236	237	231	231	235	242	238	227	219	
	Edh at Noon	43	52	63	67	64	63	59	54	46	42	39	40	
All-Sky Solar Radiation	RadAvg	589	815	1161	1469	1639	1526	1528	1586	1248	1007	712	529	
	RadStd	44	79	96	129	173	131	230	167	109	89	59	52	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+79
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page